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Rotated paleomagnetic poles from the Sassamansville syncline, Newark basin, southeastern Pennsylvania

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Abstract. A paleomagnetic fold test was conducted at eight sites collected from the Sassamansville syncline which folds the Norian age Passaic Formation along the border fault of the Newark basin in southeastern Pennsylvania. Thermal demagnetization isolated two components of magnetization; a synfolding age, intermediate unblocking temperature, normal polarity B component with intermediate inclinations and northwesterly declinations and a prefolding age, high unblocking temperature, dual polarity C component with shallow inclinations and northwesterly-southeasterly declinations. The paleomagnetic poles calculated from these magnetizations are both rotated ≈20°±6° counterclockwise from contemporary paleopoles isolated from other parts of the Newark basin. A paleomagnetic study of the Sassamansville diabase (Kodama and Mowery, 1993) indicates a 15°±5° counterclockwise rotation, so the 20° rotation of the Passaic Formation remanence at the Sassamansville syncline is interpreted to be due to a 15° block rotation of the Sassamansville area. The remainder although not statistically distinguishable may be due to grain scale strain based on rock and magnetic fabric measurements. Anisotropy of magnetic susceptibility, low temperature:high temperature susceptibility ratios, and X ray pole figure goniometry results suggest that a tectonic fabric developed in the phyllosilicates, which has vertical minimum axes and maximum extensional axes parallel to the fold axis, could have caused the small (≤5°) grain-scale rotation of the B and C component magnetizations. Fry analysis of quartz grains, however, does not record a significant finite strain in the rocks and may remain the best method to identify if a large grain-scale rotation of remanence has occurred. If local block rotations also affected some of the sites collected from the nearby Jacksonwald syncline, the effect on the B and C component paleopoles reported by Witte and Kent (1991) and Witte et al.(1991) from the Newark basin is small enough that the discrepancy between Newark basin paleopoles and coeval poles from the southwestern United States is not appreciably reduced. When the 20° rotations are removed from the Sassamansville paleomagnetic data the B component magnetization yields a high latitude paleomagnetic pole which supports the Newark B remagnetization pole of Witte and Kent (1991) and the Moat Volcanics paleopole of Van Fossen and Kent (1990). The C component magnetization yields a pole consistent with the Norian age pole from the Passaic Formation reported by Witte et al. (1989).

Introduction

Evidence is accumulating from recent paleomagnetic studies [*Van Fossen and Kent*, 1990; *Witte and Kent*, 1991; *Mowery and Kodama*, 1992] that North America's apparent polar wander path moved to high latitudes during the Middle Jurassic. This interpretation conflicts with paleomagnetic data collected from Middle Jurassic rocks in the southwestern United States which show lower-latitude apparent polar wander for North America [*May and Butler*, 1986; *Kluth et al.*, 1982]. Rotating paleomagnetic poles from other continents into a North American reference frame suggests a paleopole path halfway between these two extremes [*Van der Voo*, 1992]. Resolving this conflict is important not only to accurate determination of the paleolatitude of North America in the Middle Jurassic, with significance to the degree of allochthoneity for some southern Alaska, Cordilleran, and Coastal California tectonostratigraphic terranes but to the usefulness of paleomagnetic Euler pole (PEP) analysis [*Gordon et al.*, 1984] for apparent polar wander paths. If plates move along small circles for tens of millions of years, as initially envisioned in PEP analysis, it will have important implications for the time constants of plate driving mechanisms.

All the evidence [*Witte and Kent*, 1991;*Van Fossen and Kent*, 1990; *Mowery and Kodama*, 1992] for high-latitude Middle Jurassic poles for North America comes from secondary magnetizations which are acquired when paleohorizontal is not directly known. The Moat Volcanics pole [*Van Fossen and Kent*, 1990] is postcaldera collapse in age, but interpreted to be ancient. The Newark B component pole [*Witte and Kent*, 1991] is carried by the red siltstones and sandstones of the Mesozoic Newark basin of eastern North America. The age of this intermediate unblocking temperature magnetization was constrained by a fold test at the Jacksonwald syncline in southeastern Pennsylvania. Statistical analysis indicates that this magnetization is both posttilting (plunging) and postfolding in age, although the site mean directions tend to clus-

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ter best at partial unfolding. A secondary age for the paleopole from the Sassamansville diabase in the Newark basin is based on a synfolding magnetization which fails a baked contact test at two sites [*Mowery and Kodama*, 1992; *Kodama and Mowery*, 1993; *Kodama*, 1993; K.P. Kodama et al., Paleomagnetism of baked sedimentary rocks in the Newark and Culpeper basins: Evidence for the J1 cusp and significant late Triassic apparent polar wander from the Mesozoic basins of North America, submitted to Tectonics, 1993].

In addition to the conflict between the latitude of Middle Jurassic paleopoles from eastern North America (Newark basin and White Mountain rocks) and the southwestern United States, *Kent and Witte* [1993] point out that Triassic and Early Jurassic paleopoles from the southwestern United States are 30° west of similar age paleopoles from eastern North America. They attribute this difference to a 10°-15° clockwise rotation of the Colorado Plateau.

To provide more information about the age of the Newark B component magnetization and high latitude Middle Jurassic paleopoles for North America and to determine whether block rotations could have affected paleopoles from eastern North America, a paleomagnetic study and fold test were conducted at the Sassamansville syncline in the Newark basin of southeastern Pennsylvania.

Geologic Setting

The Sassamansville syncline is located along the Mesozoic border fault in southeastern Pennsylvania about 10 km northeast of the Jacksonwald syncline (Figure 1) and folds the Norian age (ca. 215 Ma [*Witte et al.*, 1991]) red siltstones and sandstones of the Passaic Formation. Its northwest-southeast striking fold axis is roughly perpendicular to the border fault and it is therefore designated by *Schlische* [1992] as a transverse fold. Recent structural models for the formation of these transverse structures involves along-strike variation in displacement on the border fault [*Schlische and Olsen*, 1988; *Olsen et al.*, 1989; *Schlische*, 1992].

The stratigraphic patterns of onlaps and pinchouts at the Sassamansville syncline indicate folding during sedimentation suggesting that the fold is, at least, partially a growth structure [*Schlische*, 1992] . A large diabase body intrudes the Passaic Formation in the Sassamansville syncline. Since the intrusion is generally concordant in the syncline but discordant in the adjoining anticlinal structures, the folding either preceded or was contemporaneous with the intrusion [*Schlische*, 1992]. The $^{40}Ar/^{39}Ar$ geochronology [*Sutter*, 1988] indicates that the Newark basin intrusives have a crystallization age of 201 Ma. These age relationships suggest that folding at the Sassamansville syncline continued for at least 15 m.y.

The outcrop pattern shows that the Sassamansville syncline plunges toward the northwest, but a cylindrical fit to the bedding poles of the eight sites sampled for this paleomagnetic study results in a fold axis that strikes 128° and is essentially horizontal (2° plunge to the SE). Therefore the fold test was conducted by rotating each paleomagnetic site mean direction around the local strike axis.

Methods

Eight paleomagnetic sites were collected from the Passaic Formation in the Sassamansville syncline. These sites were located inside the outcrop pattern of the Sassamansville diabase. Four sites were collected from each limb of the fold (Figure 1). From 6 to 13 individually oriented cores were

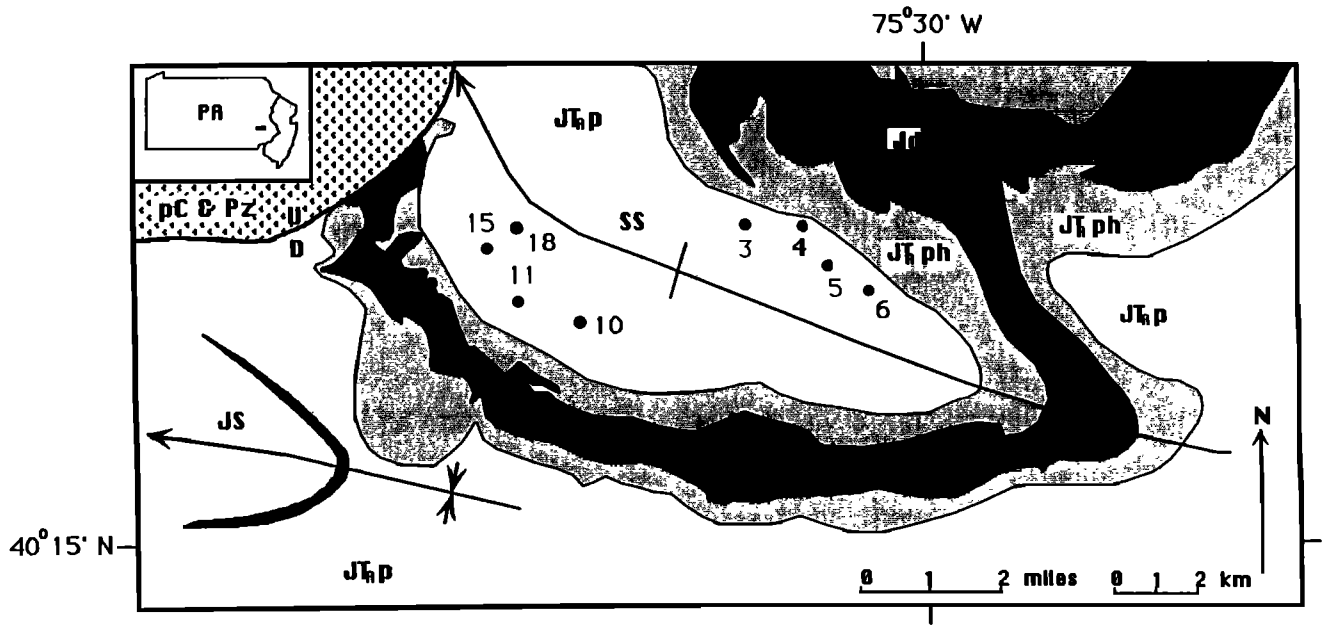


Figure 1. Generalized geologic map of the Sassamansville syncline showing its relationship to the Newark basin border fault [after *Olsen et al.*, 1989] and the location of the paleomagnetic sites sampled in this study. The dark stippled pattern indicates Mesozoic diabase, and the light stippled pattern indicates the sediments baked by the diabase. JTrp, Jurassic-Triassic Passaic Formation; Jd, Jurassic diabase; JTrph, Jurassic-Triassic Passaic Formation hornfels; JS, Jacksonwald syncline; SS, Sassamansville syncline; pC and Pz, undifferentiated Precambrian and Paleozoic rocks.

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drilled with a gasoline-powered diamond coring drill at each site. These samples were thermally demagnetized in 12 to 18 steps up to 685°C in a Schonstedt TSD-1 thermal demagnetizer and their remanence measured on a two-axis CTF superconducting magnetometer. Principal component analysis was used to isolate components of magnetization [*Kirschvink*, 1980]. The statistical significance of the fold test was calculated using the techniques of *McElhinny* [1964], *McFadden and Jones* [1981], and *McFadden* [1990].

To determine if grain-scale strain had affected the remanence, the fabric of the magnetic grains, quartz grains, and phyllosilicate grains of representative samples was measured. The anisotropy of magnetic susceptibility (AMS) of 11 samples from four sites (3, 5, 11, and 15) was measured to quantify the magnetic fabric. These 11 samples had not been thermally demagnetized, were the optimal length/diameter ratio to minimize sample shape effects on the AMS results, and had not been cemented to repair breakage which had occurred during sampling. A Sapphire SI-2 susceptibility meter was used for these measurements. To determine the proportion of the susceptibility carried by paramagnetic minerals, the bulk susceptibility of these 11 samples was measured at room temperature and at liquid nitrogen temperature (77°K.) [*Schultz-Krutisch and Heller*, 1985]. To quantify the fabric in the quartz grains, four oriented thin sections were cut from the four sites from which AMS data were acquired. These thin sections were cut parallel to the present-day horizontal, based on the AMS fabric. Finite strain was measured using center-to-center and object strain techniques [*Ramsay*, 1967; *Dunnet*, 1969; *Fry*, 1979; *Erslev*, 1989]. In each thin section the outlines of from 125 to 212 quartz grains were digitized. From these grain distributions and shapes a best fit ellipse was calculated from the enhanced normalized Fry diagram [*Erslev*, 1988]. To quantify the phyllosilicate fabric, two perpendicular thick sections (100 μm thick) were cut from each of the four samples used for Fry analysis and measured on Cornell University's Scintag X ray pole figure goniometer in transmission mode. The samples were scanned at a 2θ of 9° (illite/muscovite (002) reflection) with background intensities counted at 2θs of 10.5° and 17°. The best fit principal strain magnitudes were calculated assuming constant volume strain and grain rotation following the March [1932] model; (see review by *Oertel* [1983]).

Results

Thermal demagnetization (Figure 2) isolated an intermediate, distributed unblocking temperature magnetization (B component) between 200° and 645°C at six sites. A seventh site had unblocking temperatures as high as 680°C for this magnetization. The magnetization is normal polarity and has a northwesterly and intermediate down direction. It is the characteristic magnetization at three sites from the northern limb of the fold (3, 4, and 6; Table 1). A high unblocking temperature characteristic magnetization (C component) was isolated at five sites (5, 10, 11, 15, and 18; Table 1). It has a narrow unblocking temperature range from 650° to 685° and shallow inclinations. The magnetization is reversed polarity at four sites from the southern limb of the fold and is normal polarity at one site from the northern limb. Its declinations are southeast-northwest. This low inclination magnetization was not present in three of the sites from the north limb. Two sites had unblocking temperatures only as high as 645°C, the third site had a univectorial, normal polarity, intermediate inclination

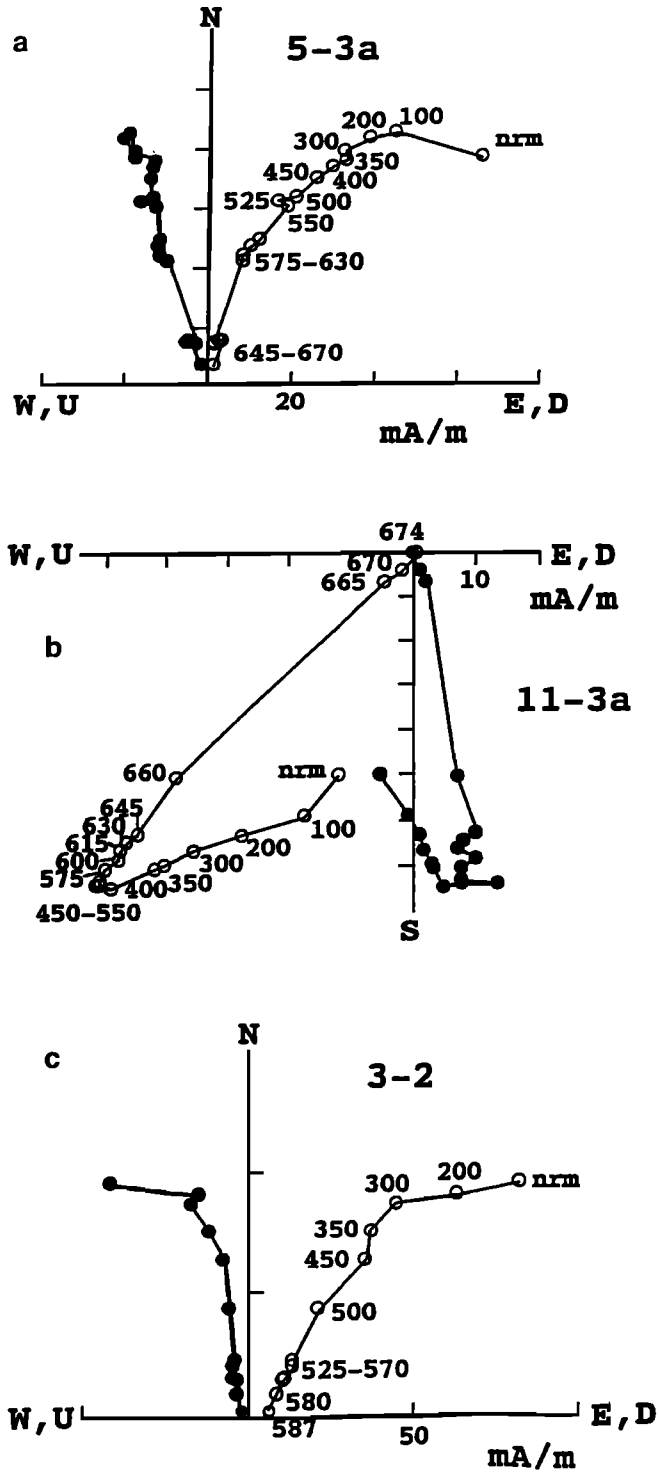


Figure 2. Orthogonal vector end point diagrams [*Zijderveld*, 1967] indicating the behavior of representative samples during thermal demagnetization. (a) Sample 5-3a shows a high temperature, normal polarity C component magnetization. (b) Sample 11-3a shows normal polarity B component and reversed polarity C component magnetizations. (c) Sample 3-2 shows a sample that only contains the normal polarity B component magnetization. Open symbols indicate projection on a vertical plane, solid symbols indicate projection on a horizontal plane.

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Table 1. B and C Component Site Mean Directions From Passaic Formation at the Sassamansville Syncline (Geographic Coordinates)

Site	N	D/DD, deg.	B Magnetization				C Magnetization			
			N_B	I/D, deg.	k_B	α_{95B}, deg.	N_C	I/D, deg.	k_C	α_{95C}, deg.
3	7	33/226	6	25.6/358.4	135.5	4.9	-	-	-	-
4	7	24/202	5	24.1/342.1	122.9	5.7	-	-	-	-
5	8	27/200	6	26.7/340.1	103.4	5.6	7	5.7/344.9	37.6	8.6
6	6	34/222	6	29.0/334.3	138.6	4.9	-	-	-	-
10	6	28/33	6	51.7/345.7	90.0	6.0	6	-37.9/158.7	260.9	3.5
11	13	29/36	13	50.8/340.5	28.3	7.3	13	-40.4/162.3	49.9	5.5
15	9	26/70	9	40.9/337.3	122.4	4.2	5	-19.2/156.7	114.3	5.9
18	6	21/32	-	-	-	-	6	-17.0/162.5	120.7	5.2

N, number of samples; D/DD, dip/dip direction of bedding; N_B or N_C, number of samples for B component or C component; I/D, Inclination/Declination; k_B or k_C, precision parameter for B or C component; α_{95B} or α_{95C}, 95% cone of confidence.

magnetization present to 680°C. This magnetization was similar in direction to the B component magnetization isolated at sites which contained both B and C components. It was interpreted to be a B component magnetization based on its distributed range of unblocking temperatures and its direction. The unblocking temperature spectra of each magnetization is consistent with hematite as the magnetic carrier in these rocks.

A stepwise, partial unfolding of the site mean directions indicates that the B component magnetization is a synfolding age magnetization (Figure 3). The site mean directions show maximum clustering at 45% unfolding which is statistically distinguishable from both the 0% and 100% unfolding distributions at the 99% significance level using both the *McFadden and Jones* [1981] and *McFadden* [1990] statistical tests. A paleomagnetic pole calculated from the average of the

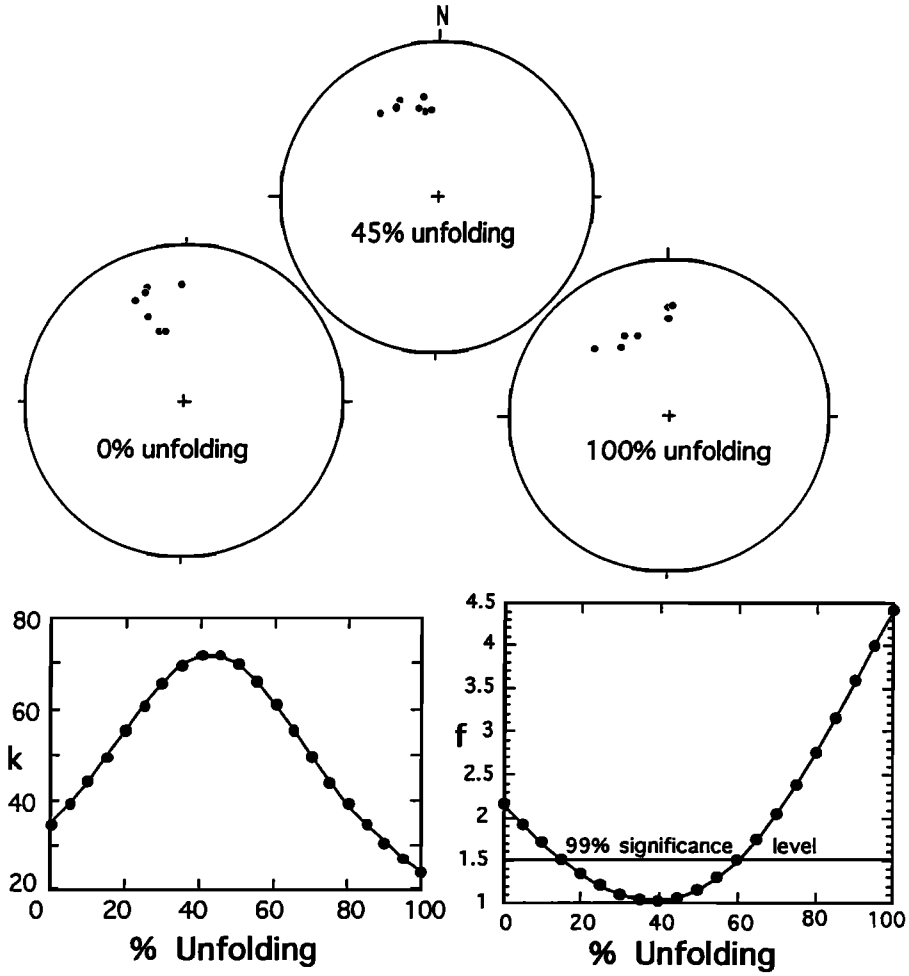


Figure 3. Stereonet plots of site mean directions and the variation in k, the precision parameter, and f, *McFadden and Jones*' [1981] fold test statistic, during stepwise unfolding of the B component magnetization. The B component magnetization shows the best clustering at 45% unfolding. Solid symbols are lower hemisphere projections.

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Table 2. Site Mean VGPs for B and C Magnetizations

Site	B Magnetization VGPs (45% Unfolding)			C Magnetization VGPs (70% Unfolding)		
	Latitude	Longitude	dp, dm	Latitude	Longitude	dp, dm
3	67.8	125.8	3.3,5.6	-	-	-
4	60.5	150.2	3.6,6.4	-	-	-
5	60.4	157.9	3.8,6.5	56.6	137.5	4.8,9.0
6	53.1	268.6	3.2,5.6	-	-	-
10	73.7	120.8	4.6,7.4	60.9	129.0	2.0,3.8
11	76.2	132.1	5.5,9.0	63.1	120.6	3.2,6.0
15	69.7	139.8	3.1,5.1	56.1	135.3	3.2,6.2
18	-	-	-	50.9	128.9	2.6,5.2

dp, dm, semiaxes of the 95% ellipse of confidence parallel (dp) or perpendicular (dm) to the site-to-pole great circle; latitude, longitude, and dp, dm in degrees.

B component site mean VGPs at 45% unfolding is located at 66.8°N, 146.5°E (A95=7.1°) (Table 2). When it is compared to *Witte and Kent's* [1991] B component paleomagnetic pole for the Newark basin, it suggests that the Sassamansville fold has been rotated 19.1° (±6.4°) counterclockwise with respect to the rest of the basin. Poleward translation is not statistically significant (2.6°±5.9°) [*Demerest*, 1983].

The C component site mean directions show a maximum clustering at 70% unfolding (Figure 4). This distribution is statistically distinguishable at the 95% significance level from the 0% unfolding distribution, but not at the 100% distribution [*McFadden*, 1990], so the magnetization is interpreted to be prefolding in age. The *McFadden and Jones* [1981] statistical test was not applicable in this case since the C component is isolated at only one site from the northern limb of the fold. When the one normal polarity site is compared to the mean direction of the four reversed polarity sites at 70% unfolding, the directions differ by 4.2° suggesting that

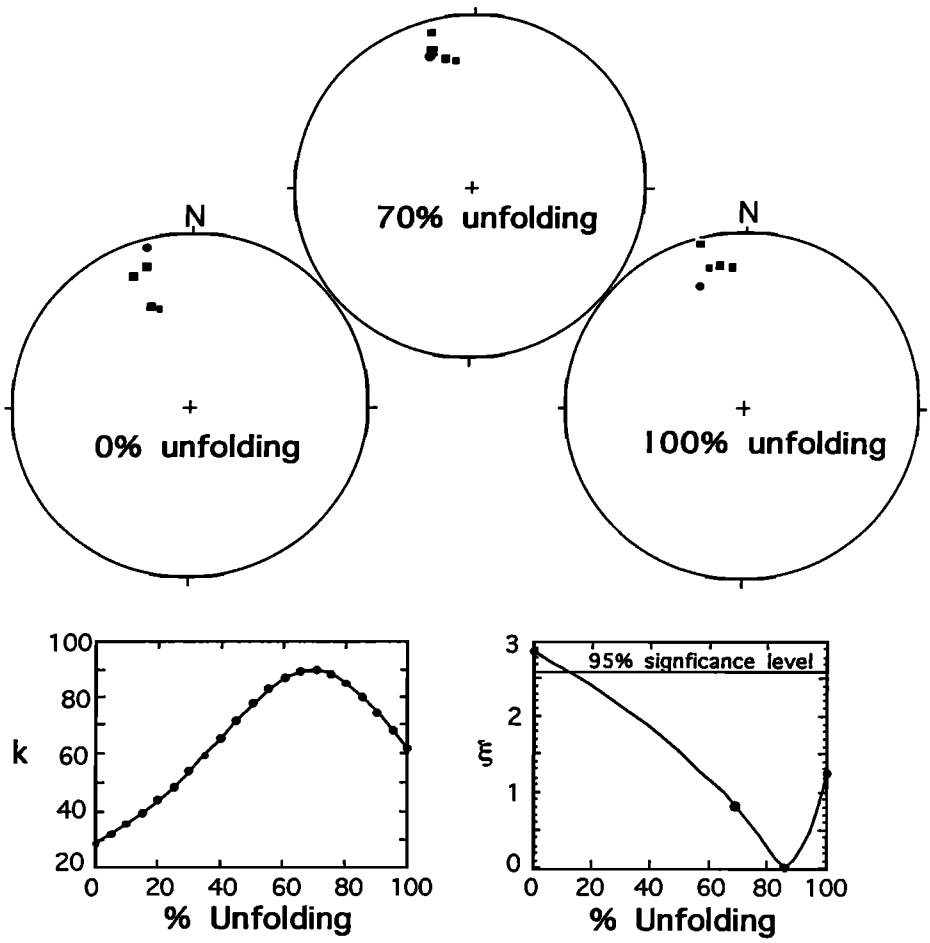


Figure 4. Stereonet plots of site mean directons and the variation in k, the precision parameter, and ξ, *McFadden's* [1990] fold test statistic, during stepwise unfolding of the C component magnetization. The solid points in the ξ versus percent unfolding plot show the calculated values for ξ. The lines between these points have been interpolated. Solid squares indicate reversed polarity site means that have been projected through the origin. The C component magnetization shows best clustering at 70% unfolding. Solid symbols are lower hemisphere projections.

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they are antipodal. *McFadden and McElhinny's* [1990] statistical test, however, classifies the reversal test as indeterminate since the critical angle, γ_c, is 24.5°. If the paleomagnetic pole calculated from the mean of the C component site mean VGPs at 70% unfolding (57.6°N, 130.6°E, A_{95}=4.5°) (Table 2) is compared to *Witte et al.'s* [1991] Norian paleomagnetic pole from the Newark basin, this comparison indicates an 19.2° (±5.7°) counterclockwise rotation of the area and a statistically indistinguishable poleward translation (-3.7°±5.7°) [*Demerest*, 1983].

The principal axes of the AMS ellipsoids show the best clustering in geographic coordinates (Figure 5a) suggesting a tectonic, rather than a depositional, origin of the fabric. The minimum axes are vertical, the maximum axes are parallel to the fold axis, and the intermediate axes are horizontal and perpendicular to the fold axis. The SW-NE trending axes (eight intermediate and three maximum) show a statistically significant clustering at 0% unfolding [*McElhinny*, 1964]. As expected

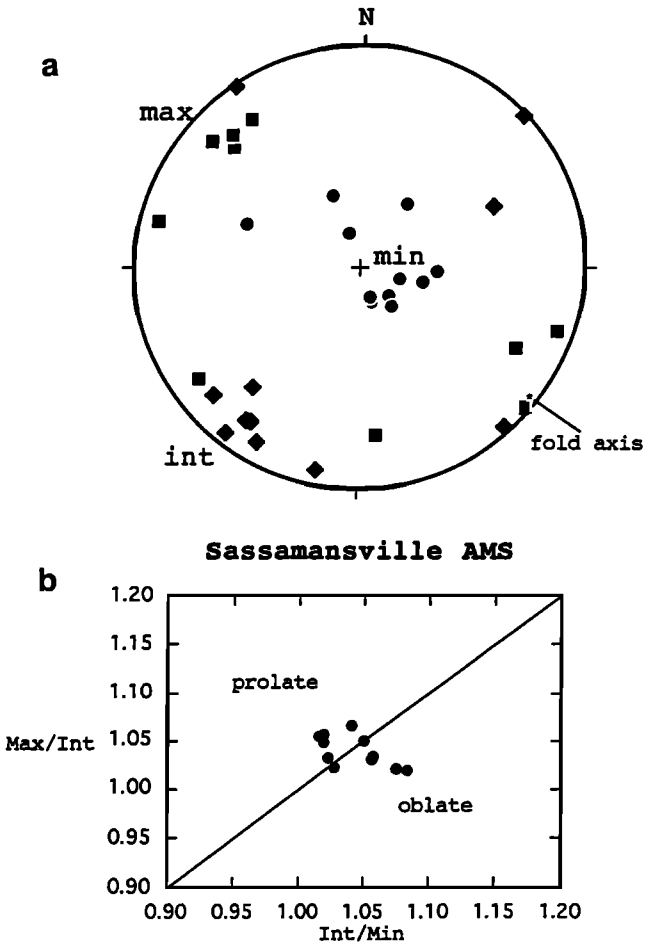


Figure 5. (a) Stereonet plot of the principal axes of the anisotropy of magnetic susceptibility (AMS) measured at sites from the northern and southern limbs of the Sassamansville syncline. The directions are plotted in geographic coordinates, solid symbols are lower hemisphere projections. Minimum axes (circles) cluster around the vertical, intermediate axes (diamonds) cluster in the horizontal, perpendicular to the fold axis (star), and maximum axes (squares) cluster in the horizontal parallel to the fold axis. (b) Ratios of AMS principal axes plotted on a Flinn [1965] diagram. The AMS axial ratios indicate a triaxial fabric.

from the distribution of principal axes a plot of the axial ratios on a Flinn diagram (Figure 5b) suggests a triaxial fabric with percent anisotropies ((maximum-minimum/intermediate)x100) averaging 7.1%.

The finite strain recorded by quartz grains [*Fry*, 1979; *Erslev*, 1989] did not correspond to the magnetic fabric measured by AMS. Axial ratios of the best fit strain ellipse were low (Table 3), and there was no consistent orientation of the maximum extensional axes (Figure 6). However, the finite strain analyses yielded poorly to moderately well-defined normalized Fry vacancy fields and errors were moderately high, possibly because of either nonuniform initial quartz grain distributions or a large fraction of phyllosilicate grains (>50%) in the rocks.

Table 3. Enhanced Normalized Fry Analysis Results

SITE	X/Y	ϕ, deg.	Average Error, %
3	1.291	215.2	25
5	1.024	7.5	24
11	1.184	3.0	25
15	1.292	320.7	20

X/Y is the axial ratio of the strain ellipse; ϕ is the angle in the horizontal plane with true north.

The best X ray pole figure goniometry results came from sites 3 and 15. In these two sites the observed intensity distribution due to the preferred orientation of illite/muscovite particles can be fit reasonably well by a theoretical intensity distribution based on *March* [1932] rotation of grains during strain (Figure 7 and Table 4). Sites 5 and 11 do not provide as good a fit, probably due to air scatter (J. Morgan, written communication, 1993). At sites 5, 11, and 15 the minimum principal axes are vertical, similar to the AMS fabric, and the maximum and intermediate principal axes are horizontal. The maximum axes are scattered about the fold axis with the intermediate axes roughly perpendicular to the fold axis. The best fit principal strain magnitude (March strains, assuming constant volume strain) is the greatest for site 15. The orientation of site 15's principal axes are in very good agreement of the AMS principal axes.

The ratio of susceptibility measured for samples cooled to liquid nitrogen temperature (77°K) to susceptibility measured at room temperature (294°K) ranges from an average of 1.25 for site 15 to an average of 1.93 for site 11 (Table 5). Based on the Curie law of paramagnetic susceptibility this ratio should be 3.8 for a rock in which only paramagnetic grains contribute to the susceptibility. The low-temperature susceptibility results suggest that about half the susceptibility at site 11 is due to paramagnetic minerals (and half due to ferromagnetic hematite) while site 15's susceptibility is due to approximately a 1:2 ratio of paramagnetic to ferromagnetic contributions. Interestingly, the site with the lowest paramagnetic contribution (15) has the most scatter in its AMS principal axes while the site with the highest paramagnetic contribution (11) shows the best clustering of its AMS principal axes (Figure 8).

Discussion

The paleomagnetic poles calculated from the B and C magnetization components isolated in the Passaic Formation at the Sassamansville syncline are significantly displaced from

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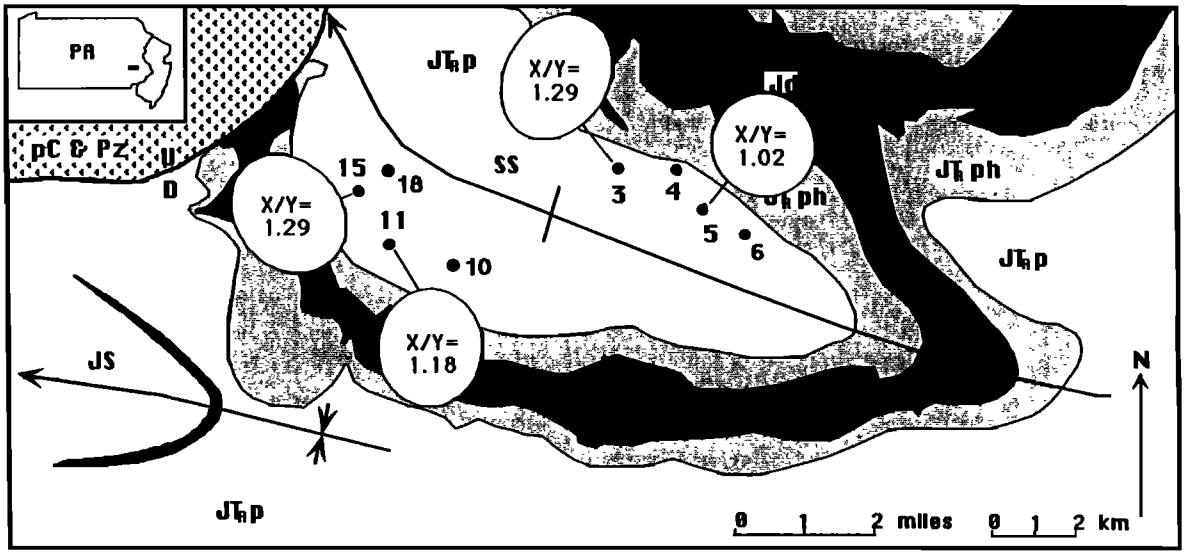


Figure 6. Generalized geologic map (same as Figure 1) showing the finite strain ellipses determined by *Fry* [1979] analysis of quartz grains in oriented thin sections collected at four sites from the northern and southern limbs of the Sassamansville syncline. The strain ellipses lie in the present day horizontal and show relatively low levels of finite strain with no preferred orientation of the ellipses' extensional axes. Abbreviations same as for Figure 1.

contemporary paleopoles derived by *Witte and Kent* [1991] (B component) and *Witte et al.* [1991] (C component) from other parts of the Newark basin (Figure 9). This result could be explained by at least two different mechanisms. Either the area of the Sassamansville syncline has undergone a local counterclockwise block rotation with respect to the rest of the Newark basin or grain-scale strain has rotated the magnetic grains carrying both the B and C components of magnetization. Each explanation has geological support. Tectonic models for the Newark-Gettysburg rift system predict sinistral shear parallel to the "neck" between the Newark and Gettysburg basins [*Lucas et al.* 1988]. The Sassamansville syncline lies along the border fault adjacent and just north of this neck, hence the area may have been rotated counterclockwise by the postulated sinistral shear. Other geologic evidence in support of a local block rotation for the Sassamansville area comes from the Chalfont fault, a large, east-west trending intrabasinal fault, which terminates in the hinge of the small anticline adjacent and north of the Sassamansville syncline [see *Schlische,* 1992, Figures 9 and 10]. Since the dip of the fault is unknown, the nature of the slip is not well constrained. *Schlische* [1992] points out that the density of intrabasinal faulting is greater south of the Chalfont fault compared to north of the fault. *Schlische* therefore concludes the Chalfont is a transfer fault [*Gibbs,* 1984] separating two regions having differential extension, the block south of the fault having been extended more than the northern block. The fault dies out, according to Schlische, where the differential extension diminishes. Perhaps the displacement has been transferred to a local block rotation at the Sassamansville syncline which is located just southwest of the Chalfont fault termination.

A grain-scale strain explanation for the counterclockwise rotation of the paleomagnetic signal is also supported by geologic evidence. Both mudcracks and dinosaur footprints show evidence of a ductile deformation of Passaic Formation rocks with extension parallel to the fold axis of the Jacksonwald syncline [*Schlische*, 1992; *Olsen et al.*, 1989; *Lucas et al,* 1988]. A similarly oriented coaxial strain, with extension parallel to the fold axis of the Sassamansville syncline, would rotate the north trending B and C magnetization components observed by *Witte and Kent* [1991] and *Witte et al*. [1991] in the other parts of the Newark basin to the northwest at the Sassamansville fold. Our AMS results would support the existence of this fabric in the Sassamansville area.

Each interpretation is significant. Local block rotations in the Newark basin could help explain the 10°-15° discrepancy between late Triassic results from the southwestern United States and the Newark basin [*Kent and Witte,* 1993]. Alternatively, if grain-scale strain is the best explanation, then Fry analysis may not always be the best method to determine if grain scale strain has affected the accuracy of the paleomagnetic signal, a technique favored by *Stamatakos and Kodama* [1991a, b] in their work on Paleozoic redbeds in the Appalachians. It would also suggest that, based on the different unblocking temperature spectra of the B and C component magnetizations, hematite grains with different sizes and grain size distributions, have behaved very similarly during deformation.

Kodama and Mowery [1993] have recently reported on a paleomagnetic study of the diabase which intrudes the Passaic Formation at the Sassamansville syncline. They find that the diabase has a statistically significant [*McFadden and Jones,* 1981] synfolding magnetization. When this secondary magnetization is compared to the B component remagnetization that *Witte and Kent* [1991] have isolated from other parts of the Newark basin, a 15.1° (±5.2°) counterclockwise rotation for the Sassamansville diabase is indicated. A 15° counterclockwise rotation for the diabase is consistent but not as large as the nearly 20° counterclockwise rotation observed for the B and C component magnetizations isolated from the Passaic Formation sedimentary rocks. The confidence intervals on the rotations for the diabase and sedimentary rocks indicate that the rotations are not statistically distinguishable; however, the diabase is a more competent unit than the sedi-

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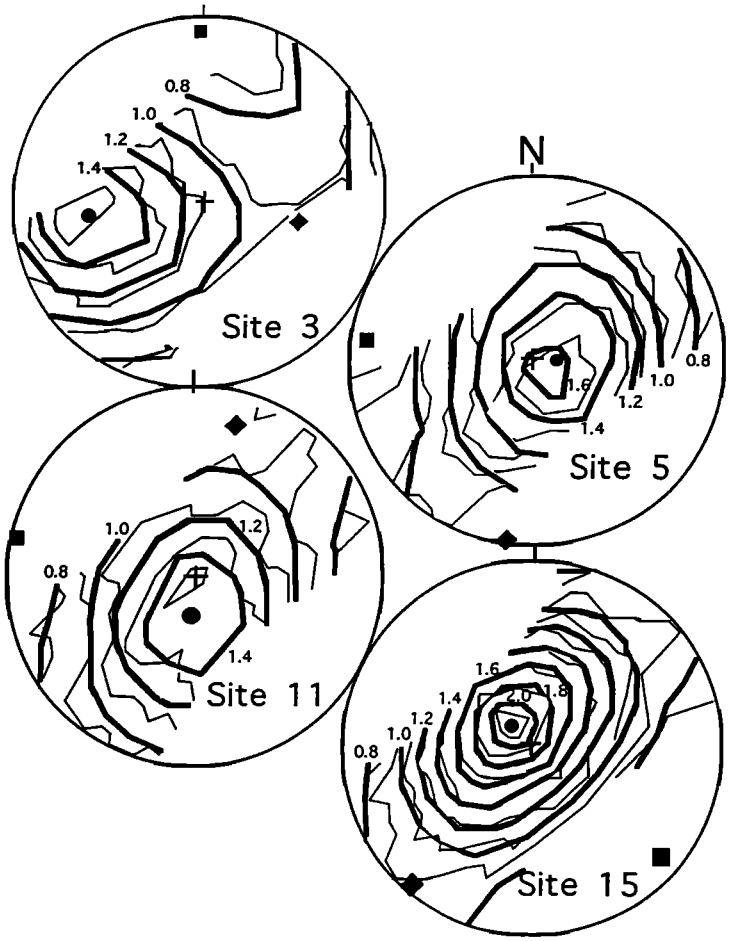


Figure 7. Stereonets showing X ray pole figure goniometry results from the four sites which yielded the Fry finite strain results. The thin lines are the contours of the X ray (002) reflections from illite/muscovite grains. The bold lines represent the theoretical intensity distribution that would result from the principal strains fit to the pole figure data assuming constant volume deformation and grain rotation following the March model. The solid circles, diamonds, and squares represent the minimum, intermediate, and maximum principal strain axes, respectively. The minimum axes at three sites cluster around the vertical, similar to the AMS results. The maximum axes lie near to the horizontal and roughly cluster in a northwest-southeast direction, again similar to the AMS results. Site 15, which has the highest strain measured by the X ray pole figure technique and shows a good fit between the measured and best fit reflection distributions, has principal strain axes that closely fit the AMS results. Contours are in multiples of a random distribution (mrd).

Table 4. Principal March Strains From X Ray Pole Figure Goniometry

	Principal Axes	Orientations		Multiples of a Random Distribution	χ^2
		D, deg.	I, deg.		
Site 3					1.753
Maximum	1.137	001	11	0.68	
Intermediate	1.030	103	46	0.92	
Minimum	0.854	261	42	1.61	
Site 5					2.675
Maximum	1.141	277	11	0.67	
Intermediate	1.050	187	02	0.86	
Minimum	0.835	087	79	1.72	
Site 11					2.639
Maximum	1.138	282	02	0.68	
Intermediate	1.030	013	15	0.92	
Minumum	0.854	186	74	1.61	
Site 15					2.033
Maximum	1.225	129	12	0.54	
Intermediate	1.058	219	02	0.84	
Minimum	0.771	315	78	2.18	

χ^2 is a measure of the fit between the observed pole figure and the theoretical distribution calculated assuming a *March* [1932] rotation of the phyllosilicate particles.

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Table 5. Susceptibility Results at 77° and 294°K

Sample	Susceptibility, $4\pi \times 10^{-5}$ SI		Ratio,
	294°K	77°K	$\chi(77°):\chi(294°)$
3-5B	2.14	3.65	1.71
3-7A	2.01	2.01	1.73
3-8B	2.11	3.34	1.59
5-3B	1.35	2.24	1.66
11-1B	1.38	2.46	1.78
11-4B	1.41	2.82	1.99
11-8B	1.44	2.80	1.94
11-10B	1.33	2.65	1.99
15-3A	2.42	2.93	1.21
15-5A	2.06	2.67	1.29
15-6B	2.16	2.70	1.25

mentary rocks of the Passaic Formation and it is reasonable to assume that it has not been affected by any internal deformation. This assumption is supported by the observation that the intensity of magnetic fabric, measured by anisotropy of anhysteretic remanence (AAR), shows no relationship to site mean declinations in the diabase [*Kodama and Mowery*, 1993]. There is, however, clear geologic and AMS evidence of an internal deformation of the sedimentary rocks at the Sassmansville syncline with a horizontal extension parallel to the fold axis. The best explanation may be that the nearly 20° counterclockwise rotation recorded by the B and C components of the Passaic Formation may be due to about 15° of local block rotation and about 5° of grain-scale strain.

If a 15° counterclockwise local block rotation has affected the Sassamansville area, is it possible that block rotations could have affected other results from the Newark basin? The paleopole calculated from *Witte and Kent*'s [1991] B component remagnetization sites from outside the Jacksonwald syncline (71.1°N, 100.1°E, A95=2.5°(site means corrected for 1/3 bedding dip)) indicates a 7.5° (±6.3°) counterclockwise rotation for the B component sites collected from the Jacksonwald syncline (palepole = 79.2°N, 133.0°E, A95=6.6°(site means corrected for 1/3 folding, 1/3 plunge)). Similarly, when the paleopoles calculated from the C component magnetizations [*Witte et al.*, 1991] are divided into Norian age sites located within (paleopole = 56.8°N, 99.1°E, A95=4.6°) and outside of (paleopole = 54.0°N, 88.3°E, A95=6.2°) the Jacksonwald syncline, a 6.5° (±6.1°) counterclockwise rotation of the Jacksonwald syncline with respect to other Newark basin locations is indicated. These rotations are barely significant at the 95% confidence limit, but the mean rotations are about one half of the rotation observed at the Sassamansville syncline 10 km to the northeast. This would indicate that block rotations along the Newark basin border fault may be more common than previously thought. Interestingly, the C component site VGPs (virtual geomagnetic poles) from the Jacksonwald syncline appear to be scattered along a small circle with the easternmost Jacksonwald sites (*Witte et al.*'s [1991] sites TJM-TJR), those closest to the Sassamansville syncline, hav-

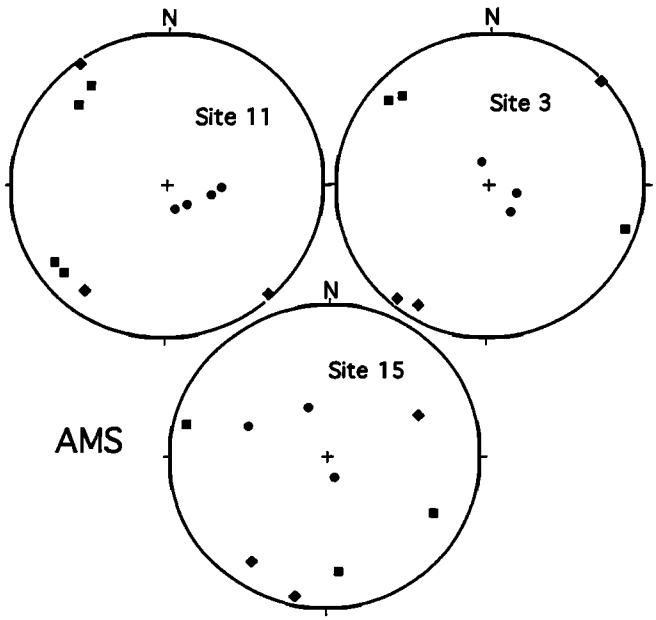


Figure 8. Stereonet plots showing the clustering of AMS principal axes at sites 3, 11, and 15. The low-temperature susceptibility results indicate that paramagnetic minerals (phyllosilicates) contribute the most to the susceptibility measured at site 11 and this site shows the best clustering of its AMS principal axes. Site 15 has the lowest contribution of paramagnetic minerals to the susceptibility and shows the most scatter in the AMS principal axes. All points are lower hemisphere projections.

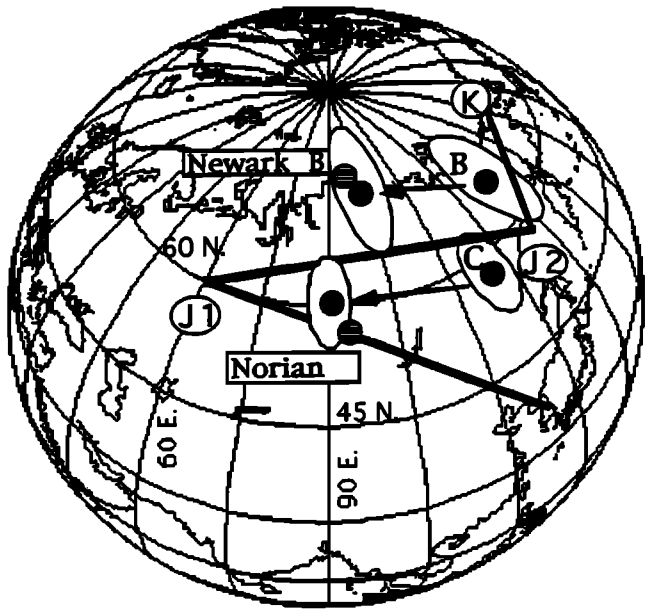


Figure 9. Paleomagnetic poles determined from the B and C component magnetizations of the Sassamansville syncline. The solid lines indicate the North American apparent polar wander path determined by *May and Butler* [1986] using what they judged to be the most reliable individual Jurassic paleopoles. Solid circles indicate paleopoles calculated from the mean of the B and C component site means at maximum K. Also shown is their 95% confidence intervals. Note that these poles fall far from the Norian and Middle Jurassic portions of the North American pole path. The stippled circles are the B and C component pole positions when these poles are rotated clockwise by 19°. The rotated B component pole comes into close agreement with *Witte and Kent*'s [1991] Newark B remagnetization pole (horizontal line pattern) and the rotated C component pole comes into close agreement with *Witte et al.*'s [1991] Norian paleopole (horizontal line pattern) for the Newark basin.

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ing the largest rotations. This could indicate relative rotations occurring between sites within the Jacksonwald syncline. It may also indicate that the largest rotations are centered on the Sassamansville area. Since a large proportion of the sites used to calculate the Newark basin B and C component reference paleopoles come from the Jacksonwald area (30% for the B component; 64% for the C component), it is probably best to use the B and C paleopoles calculated from non-Jacksonwald sites as reference poles (see paleopoles above). However, it should be noted that the angular differences between the B and C paleopoles reported by *Witte and Kent* [1991] and *Witte et al.* [1991] and the B and C paleopoles with the Jacksonwald sites removed are very small (3.1° and 4.4°, respectively). Hence this type of filtering of paleomagnetic data from the Norian strata of the Newark basin has little effect on the overall disagreement with similar age paleopoles from the southwestern United States.

Grain-scale strain could be the explanation for, at most, 5° of the remanence rotation in the Passaic Formation redbeds. For these small rotations, Fry analysis of quartz grains does not record a significant finite strain in these rocks. It is the AMS, with support from the X ray pole figure goniometry data, which indicates the presence of a tectonic fabric. The consideration of both the low-temperature susceptibility data (Table 5) and the AMS results suggests that paramagnetic minerals may be controlling the AMS fabric even though the ferromagnetic and paramagnetic minerals appear to contribute about equally to the overall susceptibility. The low-temperature to room temperature susceptibility ratios at site 11 show the highest contribution of paramagnetic minerals to the susceptibility measured at this site and the AMS fabric is well-defined (Figure 8). At site 15, however, where the low temperature to room temperature susceptibility ratios indicate the lowest paramagnetic contribution to the susceptibility, the AMS fabric is less well-defined (Figure 8), but the X ray pole figure goniometry indicates a strong phyllosilicate fabric for this site (Figure 7). Therefore the AMS fabric may not indicate significant rotation of hematite grains. This suggests that when Fry analysis indicates no discernible finite strain in a rock, this may be the best evidence that grain scale strain has not significantly affected the remanence even if AMS or pole figure goniometry results indicate a tectonic fabric.

Conclusions

The main result of this study is that the B and C component magnetizations isolated from the Passaic Formation at the Sassamansville syncline indicate a nearly 20° counterclockwise rotation of the paleomagnetic directions when compared to similar age results from other parts of the Newark basin. This remanence rotation is interpreted to be due to a combination of a 15° local block rotation and, at most, 5° caused by grain-scale strain. This result suggests that block rotations may be more common than previously thought for Newark basin sites adjacent to the border fault and that previously reported Norian [*Witte et al.*, 1991] and Middle Jurassic paleopoles [*Witte and Kent*, 1991] reported from the Newark basin may be affected by a small block rotation of the Jacksonwald syncline. However, removal of these differential rotations within the Newark basin doesn't bring coeval paleopoles from the southwestern United States and the Newark basin into agreement, so that rotation of the Colorado Plateau may be one possible explanation of the discrepancy [*Kent and Witte*, 1993].

If the 20° counterclockwise rotation is removed from the Sassamansville B and C component magnetizations, the paleomagnetic pole calculated from the C component supports the Norian paleomagnetic pole reported by *Witte et al.* [1991]. The B component pole supports secondary high latitude poles for the Middle Jurassic by *Witte and Kent* [1991] ,*Van Fossen and Kent* [1990], and *Kodama and Mowery* [1993]. The Sassamansville B component pole is synfolding, rather than the postfolding magnetization reported from the Jacksonwald syncline [*Witte and Kent*, 1991], which demonstrates that the magnetization is secondary, but definitely ancient. If the interpretation that the secondary, high latitude poles from the Newark basin is related to a hydrothermal event suggested and dated by *Sutter* [1988] to be 175-180 Ma [*Witte and Kent*, 1991], then this suggests that folding of the Sassamansville syncline continued for at least 35-40 m.y., from the Norian (circa 215 Ma, sedimentalogical evidence of synfolding deposition) to the Middle Jurassic (175-180 Ma).

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